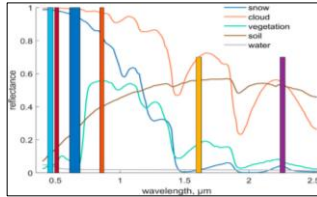


Introduction

The systematic use of Earth observation data in environmental engineering has become a key research topic. Beyond simple AI predictions, Hybrid Modeling 'replacing complex or inaccurate physical model components with AI' is gaining importance.

Reflectance 1998), spectral of water cloud, ice cloud, snow surface(Gao et al.)



Cloud contamination significantly impairs the usability of optical satellite imagery, affecting critical applications such as **environmental monitoring**, disaster response, and landuse analysis.

Research Goal

- Reconstruct cloud-contaminated optical satellite images using deep learning-based approaches to recover surface information obscured by clouds.
- Apply water body detection algorithms to both original and cloud-removed imagery to assess the improvement in detection accuracy.
- Evaluate the robustness of water body detection under varying cloud coverage conditions, quantifying how cloud removal enhances the reliability of environmental monitoring applications.

Datasets

Name	Type	Coverage	Temporal Res	Spatial Res.
Sentinel-2	Optical (Multispectral)	Global (ROIs)	5 days	10 m
Sentinel-1	SAR (Microwave)	Global (ROIs)	6 days	10 m
Cloud Mask	Binary Mask	-	-	10 m
Water Label	Binary Mask	-	-	10 m

Methodology

Cloud Removal: Prithvi-EO-2.0 Foundation Model (NASA & IBM)

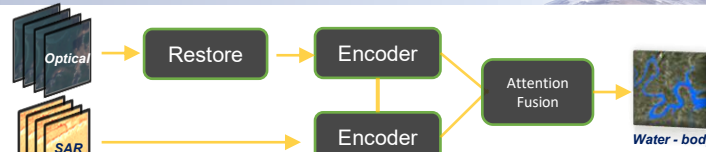
- Utilizes temporal-spatial attention mechanisms to reconstruct cloud-obscured regions
- Leverages multi-temporal inputs (4 frames) to extract clear surface information
- Learns seasonal surface variations and geographic context

Cloud-Aware Fusion

- Attention Fusion block dynamically weights optical and SAR features
- Model learns optimal modality combination based on cloud conditions
- Higher SAR weights assigned in cloud-degraded regions

Water Body Detection

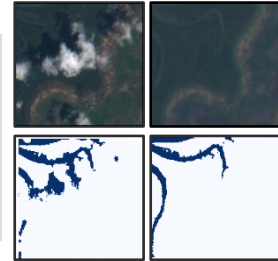
- Late fusion of restored optical (6 bands) + SAR imagery
- Robust segmentation under heavy cloud coverage
- Outperforms traditional index-based methods (MNDWI)



Results

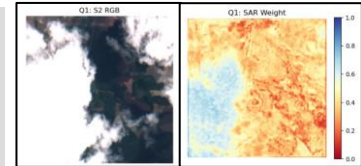
Results 1

Impact of Cloud Removal on Water Body Detection: Cloud restoration reduces false positives dramatically (1,559 → 29) and improves Water IoU by 26.7%, demonstrating that removing cloud interference is essential for accurate water segmentation.



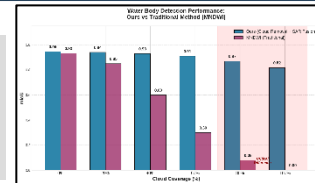
Results 2

Learned attention weights between optical and SAR modalities. The model autonomously learns to rely more on SAR where optical data is degraded by clouds.



Results 3

While MNDWI degrades significantly with increasing cloud coverage, our fusion-based model maintains stable accuracy, demonstrating superior cloud robustness.



Conclusion

- Cloud removal significantly improves water detection accuracy, with Water IoU increasing by 26.7% (0.435 → 0.551) after restoration.
- Attention Fusion effectively learns optimal modality weighting: the model automatically relies more on SAR features in cloud-covered regions, demonstrating adaptive multimodal integration.
- Our approach maintains robust performance under heavy cloud conditions, while traditional methods (MNDWI) show substantial degradation.
- Prithvi-EO-2.0's temporal attention successfully reconstructs cloud-obscured surfaces by leveraging clear observations from different time points.

Future Work :

- Optical-to-SAR translation using generative models (e.g., TerraMind) to address temporal gaps between SAR acquisitions, achieving near-daily water monitoring capability.